Akash Siddappa Goundi

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Summary

Information Science undergraduate with hands-on experience building full-stack applications and backend services using React, FastAPI, Node.js, and relational and NoSQL databases. Experienced in REST API development, system integration, real-time workflows, and integrating AI/ML components into software applications.

Experience

Software Development Engineer Intern **June 2026 – Sep 2026**
STON Technology, Bengaluru, India

- Developed backend features using **Node.js, Express.js, and MongoDB** for placement and student management workflows.
- Built QR-based admit-card verification and attendance workflows, including camera-based gate check-in and walk-in candidate registration, with persistent attendance tracking.
- Implemented CSV-based student whitelisting, bulk onboarding, join-code account claiming, **Google OAuth**, role-based access control, and **TOTP-based two-factor authentication**.
- Generated per-drive attendance reports and Excel exports with verification audit trails and edge-case alerts for placement operations.

Projects

Computer Vision-Based Smart Shopping Cart System – *React, FastAPI, PostgreSQL, Computer Vision* **GitHub**

- Developed a **React** frontend and **FastAPI** backend for product scanning, cart management, and automated checkout workflows.
- Designed persistent data models using **SQLAlchemy and PostgreSQL**, and implemented REST APIs and **Socket.IO** events for real-time cart updates and billing.
- Integrated **Grounding DINO, YOLO, and EasyOCR** for visual product identification and matching, achieving **94.2% accuracy** on the evaluated image dataset with **78 ms median inference latency**.

Offline-First AI Tutor Platform – *FastAPI, Docker, RAG, Raspberry Pi* **GitHub**

- Built **FastAPI** backend services for authentication, educational content distribution, synchronization, and AI-tutoring workflows across **Raspberry Pi** edge nodes.
- Integrated a quantized **Phi-2** model through **llama.cpp** using GGUF weights, keeping the model loaded across requests and streaming inference responses.
- Developed and evaluated a **Qdrant-based RAG pipeline** across 50 queries, achieving **90% HitRate@5**, **0.746 MRR**, and **546 ms p95 retrieval latency**.

SolBridge: Solar Energy Sharing Platform – *Express.js, PostgreSQL, Redis, MQTT* **GitHub**

- Developed backend workflows for peer-to-peer solar energy sharing, including producer-consumer matching, energy allocation, and transaction management.
- Integrated a dedicated **ML service** for forecasting and energy matching, using retries, exponential backoff, and a **circuit breaker** to improve service resilience.
- Designed data workflows using **PostgreSQL/TimescaleDB** for relational and time-series data and **Redis** for caching; benchmarked performance at **150 req/s** and **250 ms p95 latency**.

Technical Skills

Languages: Python, C++, C, JavaScript.
Frontend: React, Vite, Tailwind CSS.
Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, Socket.IO.
Databases: PostgreSQL, MongoDB, SQLite, Redis
AI/ML: LLM Integration, RAG, YOLO
Tools & Platforms: Docker, Git, GitHub, Linux.
Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design.

Education

R V College of Engineering **2023 – 2027**
B.E. in Information Science and Engineering CGPA: 8.37

Konnur Science PU College **2022 – 2023**
PUC – PCMCs Percentage: 95.16%

Achievements

- Solved **400+ Data Structures and Algorithms problems** across LeetCode and Codeforces.
- Winner – Technical Quiz Challenge at Nagarjuna College.
- Winner – Tech Arena Event at KLE Society Degree College Nagarbhavi.